

AI-Based War Damage Detection for Reconstruction Support in Lebanon

LebNet Tech Fellows Final Project

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1- Abstract

Lebanon has experienced repeated periods of conflict that have left widespread damage to buildings, complicating reconstruction planning and aid prioritization. This project builds an AI-based change detection system that identifies likely-damaged areas using before-and-after satellite imagery. A U-Net-based convolutional neural network was trained on the LEVIR-CD building change detection dataset (7,120 training image pairs) using an early fusion (6-channel stacked input) approach and a weighted loss function to address severe class imbalance (~95% unchanged / ~5% changed pixels). The trained model achieved an F1-score of 0.586 and IoU of 0.414 on a held-out LEVIR-CD test set. When applied to real war damage imagery from Lebanon (UNOSAT-documented 2006 Israel war on Lebanon damage assessments of Beirut and Bent Jbail), the model produced noisy, high-variance predictions, revealing a clear domain generalization gap between the US-based training distribution and real Western Asia conflict imagery. This gap is itself a central and informative finding, motivating future work in domain adaptation. A simple interactive Gradio interface was built to demonstrate the model on new image pairs.

2- Introduction

Following periods of armed conflict, humanitarian organizations and reconstruction planners need fast, reliable ways to identify which areas have sustained damage, in order to prioritize aid delivery and rebuilding efforts. Manually reviewing satellite imagery to assess damage across large regions is slow and does not scale with the pace at which decisions are often needed. This project addresses that problem by applying computer vision to automatically detect change between before/after satellite image pairs, as a proxy for identifying likely war-damaged areas in Lebanon.

The core goal was to build a working, end-to-end change detection pipeline: from data acquisition and preprocessing, through model training and evaluation, to testing on real Lebanese war damage imagery and presenting results through an interactive interface. A secondary goal was to honestly evaluate how well a model trained on one geographic/visual domain (available public training data) generalizes to a genuinely different one (Lebanese war damage imagery), since this generalization gap is itself directly relevant to real-world deployment of such a tool.

3- Methodology

Dataset: The original plan was to use the xBD dataset (xView2 Challenge), which provides damage-severity labels for disaster response. This was abandoned after xBD's download system proved impractical within the project timeline (signed URLs requiring browser session cookies incompatible with Colab, repeated 403 Forbidden errors). The project pivoted to LEVIR-CD, a building change detection dataset of 256x256 before/after satellite image pairs with binary change masks, accessible directly through Hugging Face's datasets library (ericyu/LEVIRCD_Cropped256; 7,120 train / 1,024 val / 2,048 test pairs). LEVIR-CD labels general urban change (including new construction) rather than war damage specifically; this

project reframes the same before/after change-detection task to the real-world problem of identifying destruction rather than construction.

Class imbalance analysis: A sample of 200 training examples showed 55% contained no labeled change at all, and changed pixels averaged only 4.93% across all samples. This directly informed the choice of evaluation metrics (F1-score and IoU rather than plain accuracy) and the use of a weighted loss function during training.

Model architecture: A simplified 2-level U-Net (encoder-decoder with skip connections) was implemented, taking a 6-channel input formed by stacking the before and after images along the channel dimension (early fusion). This approach was chosen over a Siamese architecture (two parallel encoders comparing learned features) for implementation simplicity; Siamese networks are a known stronger alternative for change detection and are discussed as future work.

Training setup: A weighted Binary Cross-Entropy loss ($\text{pos_weight} \approx 19$, derived from the class imbalance ratio) was used with the Adam optimizer (learning rate $1e-3$) for 10 epochs, on a Colab GPU runtime.

4- Implementation Details & Results

Training loss decreased consistently across all 10 epochs, from 1.1207 to 0.2081, with the steepest improvement in the first three epochs. The loss curve showed no instability or premature plateauing, and was still decreasing at epoch 10, suggesting further training would likely improve results further.

On the held-out LEVIR-CD test set (2,048 examples), the model achieved an F1-score of 0.586 and an IoU of 0.414 — both well above a naive "predict no-change everywhere" baseline (which would score near 0), indicating the model learned genuine change-detection patterns. Visual inspection of predictions confirmed this: the model reliably located general regions of change and correctly predicted "no change" for unchanged examples, though predicted regions were noisier and less precisely bounded than the clean ground-truth shapes.

Real-world testing used UNOSAT damage assessment maps for Beirut and Bent Jbail, documenting the 2006 Israel war on Lebanon (178 and 158 buildings respectively identified as destroyed/severely damaged). After cropping the pre/post-crisis panels and preprocessing to match the training pipeline, the model was run on both image pairs. Predictions were substantially noisier and more scattered than on LEVIR-CD test data, covering large, high-variance portions of both images rather than clean, localized regions.

An interactive Gradio interface was built, allowing before/after image pairs to be uploaded and the predicted change mask displayed in real time, with the Beirut and Bent Jbail pairs included as one-click example inputs.

5- Discussion & Analysis

The LEVIR-CD test results support the core hypothesis: a U-Net trained with a weighted loss function on an imbalanced change-detection dataset can learn to identify real change patterns rather than defaulting to the majority class, achieving reasonable F1/IoU scores for a first pass.

The real-world Lebanon test, however, reveals an important limitation: strong performance on a training distribution does not guarantee similar performance on visually different real-world data, even when the underlying task is conceptually identical. The noisy predictions on UNOSAT imagery likely stem from several compounding domain-shift factors: LEVIR-CD's US-based suburban/rural construction scenes differ substantially from the dense Western Asia urban architecture in the UNOSAT maps; building materials, rooftop colors, and urban density all differ from the training distribution; and the UNOSAT post-crisis images include UN-added red annotation markers overlaid directly on the imagery, a visual pattern the model never encountered during training and may be reacting to as noise. Additionally, the UNOSAT imagery documents the 2006 Israel war on Lebanon rather than more recent conflict, since it was the most accessible source of high-resolution (1m), professionally verified before/after war damage imagery; the broader pipeline is intended to generalize to other conflict periods as suitable imagery becomes available.

Key limitations:

- 1) The early fusion architecture likely caps achievable performance below what a Siamese network would offer.
- 2) Training was capped at 10 epochs due to time constraints, while the loss curve suggests the model had not yet converged.
- 3) No ground-truth pixel-level labels exist for the Lebanon test imagery, so evaluation there is qualitative rather than metric-based.
- 4) The real-world test used historical rather than current conflict imagery.

Future work should prioritize a Siamese architecture, additional training epochs, and most importantly for real-world applicability, fine-tuning or domain adaptation using a small set of Western Asia urban imagery to close the observed domain gap.

6- Reflection on Learnings

This project's most rewarding aspect was seeing the full pipeline work end-to-end: watching training loss decrease predictably, then seeing the model correctly identify real building construction in LEVIR-CD test examples, which confirmed that the underlying concepts (weighted loss for imbalance, U-Net for segmentation, early fusion for multi-image input) were understood.

The most significant challenge was data access: the original xBD dataset proved practically inaccessible within the timeline, requiring a mid-project pivot to LEVIR-CD. Sourcing real

Lebanon war damage imagery was similarly difficult: Maxar's open data program did not cover Lebanon, and Copernicus/Sentinel imagery proved too low-resolution to be useful, before UNOSAT's historical damage assessment maps were found to be a workable, high-resolution alternative. This reinforced an important lesson: validating data accessibility early, and being willing to adapt scope pragmatically, is often as important as the modeling work itself.

A second key challenge was debugging a subtle model-loading bug during real-world testing, where a previously trained model was silently overwritten by a freshly initialized one due to cell execution order, diagnosed by systematically checking raw output probability distributions rather than only the final thresholded predictions. This was a valuable lesson in verifying assumptions at each pipeline stage rather than trusting that a working component stays working.

Overall, this project provided hands-on experience with the full lifecycle of an applied computer vision system, dataset sourcing under real constraints, addressing class imbalance, architecture tradeoffs under time pressure, and honestly confronting a genuine domain generalization failure rather than hiding it, which reflects the realistic conditions of applying AI to real-world, socially meaningful problems.